

XYRO - Addendum

Security Assessment

CertiK Assessed on Feb 25th, 2025





Certik Assessed on Feb 25th, 2025

XYRO - Addendum

The security assessment was prepared by Certik, the leader in Web3.0 security.

Executive Summary

TYPES

DeFi

ECOSYSTEM

Ethereum (ETH)

METHODS

Formal Verification, Manual Review, Static Analysis

LANGUAGE

Solidity

TIMELINE

Delivered on 02/25/2025

KEY COMPONENTS

N/A

CODEBASE

[xyro-io](#)[View All in Codebase Page](#)

COMMITTS

- [149af546e072c560f24ba6900961a5fdfe81c9a4](#)
- [5d73454485b4e1a5c42d6c1993c9e24860df4b9d](#)

[View All in Codebase Page](#)

Vulnerability Summary



3

Total Findings

3

Resolved

0

Mitigated

0

Partially Resolved

0

Acknowledged

0

Declined



0

Critical

Critical risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.



0

Major

Major risks can include centralization issues and logical errors. Under specific circumstances, these major risks can lead to loss of funds and/or control of the project.



0

Medium

Medium risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform.



2

Minor

2 Resolved



Minor risks can be any of the above, but on a smaller scale. They generally do not compromise the overall integrity of the project, but they may be less efficient than other solutions.



1

Informational

1 Resolved



Informational errors are often recommendations to improve the style of the code or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

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CON-01 : Tokens Without Fee Set Will Not Be Charged a Game Fee

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CODEBASE | XYRO - ADDENDUM

Repository

[xyro-io](#)

Commit

- [149af546e072c560f24ba6900961a5fdfe81c9a4](#)
- [5d73454485b4e1a5c42d6c1993c9e24860df4b9d](#)

AUDIT SCOPE | XYRO - ADDENDUM

6 files audited ● 3 files with Resolved findings ● 3 files without findings

ID	Repo	File	SHA256 Checksum
● BUL	xyro-io/smart-contracts	 Bullseye.sol	13119c1a367c709b5bfaca93e5f5ae307c380 188d830fc5b6e879f959a6236d
● OVO	xyro-io/smart-contracts	 OneVsOneExactPrice.sol	f506e10477c828876fd6338e2711ab905accd 72d36115d48393f1bdf984687ac
● SET	xyro-io/smart-contracts	 Setup.sol	842d69364830834b246b6debc7eb75d743fc7 a22ff7b0b503aaf97546977a93b
● BUS	xyro-io/smart-contracts	 Bullseye.sol	85664ca563be668ba93683cb54fbdfbd68ac6 bb0cc904b17cd6acd7b1d01fe65
● OVE	xyro-io/smart-contracts	 OneVsOneExactPrice.sol	369ce1bb0b71d428f1e9f41a4e1966f19a0450 bb6f0355f3c2f04343750afc9c
● SEU	xyro-io/smart-contracts	 Setup.sol	e33b8af3b2212976afc0917470e1f942a2e873 5eb71e8b164918e2b5e9fa6496

APPROACH & METHODS | XYRO - ADDENDUM

This report has been prepared for XYRO to discover issues and vulnerabilities in the source code of the XYRO - Addendum project as well as any contract dependencies that were not part of an officially recognized library. A comprehensive examination has been performed, utilizing Manual Review and Static Analysis techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.

The security assessment resulted in findings that ranged from critical to informational. We recommend addressing these findings to ensure a high level of security standards and industry practices. We suggest recommendations that could better serve the project from the security perspective:

- Testing the smart contracts against both common and uncommon attack vectors;
- Enhance general coding practices for better structures of source codes;
- Add enough unit tests to cover the possible use cases;
- Provide more comments per each function for readability, especially contracts that are verified in public;
- Provide more transparency on privileged activities once the protocol is live.

REVIEW NOTES | XYRO - ADDENDUM

Overview

The **XYRO** project coordinates a series of smart contracts aimed at various onchain games. The focus of this audit is on the changes made to the following files between commits 987438a79218d81ad7df11e8a8b01e503a9a6b37 and 149af546e072c560f24ba6900961a5fdfe81c9a4:

- Bullseye.sol
- OneVsOneExactPrice.sol
- Setup.sol

FINDINGS | XYRO - ADDENDUM



3

Total Findings

0

Critical

0

Major

0

Medium

2

Minor

1

Informational

This report has been prepared to discover issues and vulnerabilities for XYRO - Addendum. Through this audit, we have uncovered 3 issues ranging from different severity levels. Utilizing the techniques of Manual Review & Static Analysis to complement rigorous manual code reviews, we discovered the following findings:

ID	Title	Category	Severity	Status
BUL-01	Missing Check For <code>pricePrecision</code>	Incorrect Calculation	Minor	● Resolved
CON-01	Tokens Without Fee Set Will Not Be Charged A Game Fee	Design Issue	Minor	● Resolved
BUL-02	Confirmation Of <code>finalPrice</code> Not Adjusted Using <code>pricePrecision</code>	Inconsistency	Informational	● Resolved

BUL-01 | MISSING CHECK FOR `pricePrecision`

Category	Severity	Location	Status
Incorrect Calculation	Minor	Bullseye.sol (02/07-149af5): 115	Resolved

Description

`pricePrecision` needs to be checked during `startGame()` to ensure it does not conflict with `exactRange`. Specifically, it should be verified whether any price, after truncating precision with `pricePrecision`, falls outside the range of `exactRange`.

For example, if `exactRange` is $5 \cdot 10^{18}$ and the selected `pricePrecision` is 20, for the final price `P` to be in the range $[x \cdot 10^{20} + 5 \cdot 10^{18} + 1, (x + 1) \cdot 10^{20} - 5 \cdot 10^{18} - 1], x \in \mathbb{N}$, no predicted price can fall within the `exactRange`.

Recommendation

It is recommended to add a `pricePrecision` check in `startGame()` to prevent conflicts with `exactRange`.

Alleviation

[XYRO Team, 02/10/2025]:

Fixed this issue, please check this commit hash: 63d5447353d4b6a6ecd6b21d2f983575db55b209

[CertiK, 02/11/2025]:

Thank you for your update and response. The current version still doesn't enforce a size restriction between `exactRange` and `pricePrecision`. There should be a constraint:

$$exactRange \cdot 2 \geq 10^{pricePrecision}$$

[XYRO Team, 02/14/2025]:

We've added check for range, please check this commit hash: e7333a9c2934806e880d7140269987b0000ec00c

[CertiK, 02/17/2025]:

In the latest commit, the `setExactRange` checks the `exactRange` but not the input `newRange`.

```
function setExactRange(
    uint256 newRange
) public onlyRole(DEFAULT_ADMIN_ROLE) {
    require(exactRange * 2 >= 10 ** pricePrecision, "Invalid exact range");
    exactRange = newRange;
    emit NewExactRange(newRange);
}
```

[XYRO Team, 02/18/2025]:

Fixed this one, please check: acb8bcc586f1b6f0f11f4095e9ee7489ffef1d9

[Certik, 02/19/2025]:

Thank you for your update and response. In the current version, only the logic within `setExactRange()` has been modified; the corresponding checks in `startGame()` remain unchanged. The `pricePrecision` is assigned after the `require` statement, so the check does not validate the input precision.

[XYRO Team, 02/25/2025]:

The project team resolved this issue by implementing the recommendation suggested. Changes have been reflected in the commit hash: <https://github.com/xyro-io/smart-contracts/tree/5d73454485b4e1a5c42d6c1993c9e24860df4b9d/contracts>

CON-01 | TOKENS WITHOUT FEE SET WILL NOT BE CHARGED A GAME FEE

Category	Severity	Location	Status
Design Issue	Minor	OneVsOneExactPrice.sol (02/07-149af5): 517~521; Setup.sol (02/07-149af5): 445~449, 474~478, 512~516, 542~546	Resolved

Description

Compared to previous versions, the `Setup` and `OneVsOneExactPricefee` now use separate fee rates for each token, rather than applying the same rate across all tokens. However, for tokens where the fee has not been set, the default rate is 0%, instead of the previous 10% and 5%. In this case, players will not be charged the game fee during game round. Players can select an approved token without a fee set as their game token, effectively bypassing the official game fee.

Recommendation

We would like to confirm if this is the expected behavior. If not, the development team will need to restrict tokens without a set fee from being used as game token in the game.

Alleviation

[XYRO Team, 02/11/2025]:

Fixed this issue, please check this commit hash: 63d5447353d4b6a6ecd6b21d2f983575db55b209.

[Certik, 02/11/2025]:

The team has only added checks in the play functions or during game creation. However, if the privileged role updates the fee to zero before finalizing a game, the issue still occurs. We would like to confirm whether this scenario has been considered and recommend adding checks during game finalization as well.

[XYRO Team, 02/14/2025]:

The scenario you described is considered and we added a default fee of 10% in the finalization if the game token fee was somehow set to 0. Please check this commit hash: 83ce57e379686cb33682505bf50d63da41706a60.

BUL-02 | CONFIRMATION OF `finalPrice` NOT ADJUSTED USING `pricePrecision`

Category	Severity	Location	Status
Inconsistency	● Informational	Bullseye.sol (02/07-149af5): 346~349	● Resolved

Description

The user guess prices are rounded according to `pricePrecision` when invoking the play functions. However, the `finalPrice` is not adjusted using `pricePrecision` in the `finalizeGame` function, resulting in an inconsistent basis for price comparison. For example, if userA's guess price is 158, userB's guess price is 168, and `pricePrecision` is 1, the stored guess price for userA becomes 150 and for userB becomes 160. If the `finalPrice` turns out to be 156, userB would be declared the winner under the current system. This may lead to confusion and perceived unfairness among users.

Recommendation

We would like to confirm with the team whether the current design is intended.

Alleviation

[XYRO Team, 02/11/2025]:

This behaviour is correct. Prices come with 18 decimals and we are using it to cut last digits. This feature should have prevented bot user from winning games. He makes two guesses altering user's highest and lowest prices by 1. But with rounding user's guesses are prioritised as they participated earlier.

FORMAL VERIFICATION | XYRO - ADDENDUM

Formal guarantees about the behavior of smart contracts can be obtained by reasoning about properties relating to the entire contract (e.g. contract invariants) or to specific functions of the contract. Once such properties are proven to be valid, they guarantee that the contract behaves as specified by the property. As part of this audit, we applied formal verification to prove that important functions in the smart contracts adhere to their expected behaviors.

Considered Functions And Scope

In the following, we provide a description of the properties that have been used in this audit. They are grouped according to the type of contract they apply to.

Verification of contracts derived from AccessControl v4.4

We verified properties of the public interface of contracts that provide an AccessControl-v4.4 compatible API. This involves:

- The `hasRole` function, which returns `true` if an account has been granted a specific `role`.
- The `getRoleAdmin` function, which returns the admin role that controls a specific `role`.
- The `grantRole` and `revokeRole` functions, which are used for granting a `role` to an account and revoking a `role` from an `account`, respectively.
- The `renounceRole` function, which allows the calling account to revoke a `role` from itself.

The properties that were considered within the scope of this audit are as follows:

Property Name	Title
accesscontrol-renouncerole-revert-not-sender	<code>renounceRole</code> Reverts When Caller Is Not the Confirmation Address
accesscontrol-getroleadmin-succeed-always	<code>getRoleAdmin</code> Function Always Succeeds
accesscontrol-hasrole-succeed-always	<code>hasRole</code> Function Always Succeeds
accesscontrol-getroleadmin-change-state	<code>getRoleAdmin</code> Function Does Not Change State
accesscontrol-default-admin-role	AccessControl Default Admin Role Invariance
accesscontrol-grantrole-correct-role-granting	<code>grantRole</code> Correctly Grants Role
accesscontrol-hasrole-change-state	<code>hasRole</code> Function Does Not Change State
accesscontrol-renouncerole-succeed-role-renouncing	<code>renounceRole</code> Successfully Renounces Role
accesscontrol-revokerole-correct-role-revoking	<code>revokeRole</code> Correctly Revokes Role

Verification Results

For the following contracts, formal verification established that each of the properties that were in scope of this audit (see scope) are valid:

Detailed Results For Contract OneVsOneExactPrice (contracts/OneVsOneExactPrice.sol) In Commit 5d73454485b4e1a5c42d6c1993c9e24860df4b9d

Verification of contracts derived from AccessControl v4.4

Detailed Results for Function `renounceRole`

Property Name	Final Result	Remarks
accesscontrol-renouncerole-revert-not-sender	● True	
accesscontrol-renouncerole-succeed-role-renouncing	● True	

Detailed Results for Function `getRoleAdmin`

Property Name	Final Result	Remarks
accesscontrol-getroleadmin-succeed-always	● True	
accesscontrol-getroleadmin-change-state	● True	

Detailed Results for Function `hasRole`

Property Name	Final Result	Remarks
accesscontrol-hasrole-succeed-always	● True	
accesscontrol-hasrole-change-state	● True	

Detailed Results for Function `DEFAULT_ADMIN_ROLE`

Property Name	Final Result	Remarks
accesscontrol-default-admin-role	● True	

Detailed Results for Function `grantRole`

Property Name	Final Result	Remarks
accesscontrol-grantrole-correct-role-granting	● True	

Detailed Results for Function `revokeRole`

Property Name	Final Result	Remarks
accesscontrol-revokerole-correct-role-revoking	● True	

**Detailed Results For Contract Setup (contracts/Setup.sol) In Commit
5d73454485b4e1a5c42d6c1993c9e24860df4b9d****Verification of contracts derived from AccessControl v4.4**Detailed Results for Function `renounceRole`

Property Name	Final Result	Remarks
accesscontrol-renouncerole-revert-not-sender	● True	
accesscontrol-renouncerole-succeed-role-renouncing	● True	

Detailed Results for Function `getRoleAdmin`

Property Name	Final Result	Remarks
accesscontrol-getroleadmin-succeed-always	● True	
accesscontrol-getroleadmin-change-state	● True	

Detailed Results for Function `hasRole`

Property Name	Final Result	Remarks
accesscontrol-hasrole-succeed-always	● True	
accesscontrol-hasrole-change-state	● True	

Detailed Results for Function `DEFAULT_ADMIN_ROLE`

Property Name	Final Result	Remarks
accesscontrol-default-admin-role	● True	

Detailed Results for Function `grantRole`

Property Name	Final Result	Remarks
accesscontrol-grantrole-correct-role-granting	● True	

Detailed Results for Function `revokeRole`

Property Name	Final Result	Remarks
accesscontrol-revokerole-correct-role-revoking	● True	

Detailed Results For Contract `OneVsOneExactPrice` (contracts/OneVsOneExactPrice.sol) In Commit `149af546e072c560f24ba6900961a5fdfe81c9a4`

Verification of contracts derived from `AccessControl v4.4`

Detailed Results for Function `getRoleAdmin`

Property Name	Final Result	Remarks
accesscontrol-getroleadmin-succeed-always	● True	
accesscontrol-getroleadmin-change-state	● True	

Detailed Results for Function `renounceRole`

Property Name	Final Result	Remarks
accesscontrol-renouncerole-revert-not-sender	● True	
accesscontrol-renouncerole-succeed-role-renouncing	● True	

Detailed Results for Function `hasRole`

Property Name	Final Result	Remarks
accesscontrol-hasrole-succeed-always	● True	
accesscontrol-hasrole-change-state	● True	

Detailed Results for Function `grantRole`

Property Name	Final Result	Remarks
accesscontrol-grantrole-correct-role-granting	● True	

Detailed Results for Function `DEFAULT_ADMIN_ROLE`

Property Name	Final Result	Remarks
accesscontrol-default-admin-role	● True	

Detailed Results for Function `revokeRole`

Property Name	Final Result	Remarks
accesscontrol-revokerole-correct-role-revoking	● True	

**Detailed Results For Contract Setup (contracts/Setup.sol) In Commit
149af546e072c560f24ba6900961a5fdfe81c9a4****Verification of contracts derived from AccessControl v4.4**Detailed Results for Function `renounceRole`

Property Name	Final Result	Remarks
accesscontrol-renouncerole-revert-not-sender	● True	
accesscontrol-renouncerole-succeed-role-renouncing	● True	

Detailed Results for Function `getRoleAdmin`

Property Name	Final Result	Remarks
accesscontrol-getroleadmin-succeed-always	● True	
accesscontrol-getroleadmin-change-state	● True	

Detailed Results for Function `hasRole`

Property Name	Final Result	Remarks
accesscontrol-hasrole-succeed-always	● True	
accesscontrol-hasrole-change-state	● True	

Detailed Results for Function `DEFAULT_ADMIN_ROLE`

Property Name	Final Result	Remarks
accesscontrol-default-admin-role	● True	

Detailed Results for Function `grantRole`

Property Name	Final Result	Remarks
accesscontrol-grantrole-correct-role-granting	● True	

Detailed Results for Function `revokeRole`

Property Name	Final Result	Remarks
accesscontrol-revokerole-correct-role-revoking	● True	

APPENDIX | XYRO - ADDENDUM

Finding Categories

Categories	Description
Incorrect Calculation	Incorrect Calculation findings are about issues in numeric computation such as rounding errors, overflows, out-of-bounds and any computation that is not intended.
Inconsistency	Inconsistency findings refer to different parts of code that are not consistent or code that does not behave according to its specification.
Design Issue	Design Issue findings indicate general issues at the design level beyond program logic that are not covered by other finding categories.

Checksum Calculation Method

The "Checksum" field in the "Audit Scope" section is calculated as the SHA-256 (Secure Hash Algorithm 2 with digest size of 256 bits) digest of the content of each file hosted in the listed source repository under the specified commit.

The result is hexadecimal encoded and is the same as the output of the Linux "sha256sum" command against the target file.

Details on Formal Verification

Some Solidity smart contracts from this project have been formally verified. Each such contract was compiled into a mathematical model that reflects all its possible behaviors with respect to the property. The model takes into account the semantics of the Solidity instructions found in the contract. All verification results that we report are based on that model.

The following assumptions and simplifications apply to our model:

- Certain low-level calls and inline assembly are not supported and may lead to a contract not being formally verified.
- We model the semantics of the Solidity source code and not the semantics of the EVM bytecode in a compiled contract.

Formalism for property specifications

All properties are expressed in a behavioral interface specification language that CertiK has developed for Solidity, which allows us to specify the behavior of each function in terms of the contract state and its parameters and return values, as well as contract properties that are maintained by every observable state transition. Observable state transitions occur when the contract's external interface is invoked and the invocation does not revert, and when the contract's Ether balance is changed by the EVM due to another contract's "self-destruct" invocation. The specification language has the usual Boolean connectives, as well as the operator `\old` (used to denote the state of a variable before a state transition), and several types of specification clause:

Apart from the Boolean connectives and the modal operators "always" (written $\Box$) and "eventually" (written $\Diamond$), we use the following predicates to reason about the validity of atomic propositions. They are evaluated on the contract's state whenever a discrete time step occurs:

- `requires [cond]` - the condition `cond`, which refers to a function's parameters, return values, and contract state variables, must hold when a function is invoked in order for it to exhibit a specified behavior.
- `ensures [cond]` - the condition `cond`, which refers to a function's parameters, return values, and both `\old` and current contract state variables, is guaranteed to hold when a function returns if the corresponding requires condition held when it was invoked.
- `invariant [cond]` - the condition `cond`, which refers only to contract state variables, is guaranteed to hold at every observable contract state.
- `constraint [cond]` - the condition `cond`, which refers to both `\old` and current contract state variables, is guaranteed to hold at every observable contract state except for the initial state after construction (because there is no previous state); constraints are used to restrict how contract state can change over time.

Description of the Analyzed AccessControl-v4.4 Properties

Properties related to function `renounceRole`

accesscontrol-renouncerole-revert-not-sender

The `renounceRole` function must revert if the caller is not the same as `account`.

Specification:

```
reverts_when account != msg.sender;
```

accesscontrol-renouncerole-succeed-role-renouncing

After execution, `renounceRole` must ensure the caller no longer has the renounced role.

Specification:

```
ensures !hasRole(role, account);
```

Properties related to function `getRoleAdmin`

accesscontrol-getroleadmin-change-state

The `getRoleAdmin` function must not change any state variables.

Specification:

```
assignable \nothing;
```

accesscontrol-getroleadmin-succeed-always

The `getRoleAdmin` function must always succeed, assuming that its execution does not run out of gas.

Specification:

```
reverts_only_when false;
```

Properties related to function `hasRole`**accesscontrol-hasrole-change-state**

The `hasRole` function must not change any state variables.

Specification:

```
assignable \nothing;
```

accesscontrol-hasrole-succeed-always

The `hasRole` function must always succeed, assuming that its execution does not run out of gas.

Specification:

```
reverts_only_when false;
```

Properties related to function `DEFAULT_ADMIN_ROLE`**accesscontrol-default-admin-role**

The default admin role must be invariant, ensuring consistent access control management.

Specification:

```
invariant DEFAULT_ADMIN_ROLE() == 0x00;
```

Properties related to function `grantRole`**accesscontrol-grantrole-correct-role-granting**

After execution, `grantRole` must ensure the specified account has the granted role.

Specification:

```
ensures hasRole(role, account);
```

Properties related to function `revokeRole`**accesscontrol-revokerole-correct-role-revoking**

After execution, `revokeRole` must ensure the specified account no longer has the revoked role.

Specification:

```
ensures !hasRole(role, account);
```

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